day1

① LinearSearchCount

[0 3 6 9 12 ...]

key = 0

i = 0

a[0] = 0

0 == 0

comparisons = 1

i = 500 → 501 comparison

i = 999 → 1000 comparison

for -1 = all 1000 checked

return -1

② Loop Complexity

for single (n)

i=0 → c=1
i=1 → c=2
i=2 → c=3
i=3 → c=4
i=4 → c=5

so O(n)

nested (3)

i=0 j=0,1,2 → 3
i=1 j=0,1,2 → 3
i=2 j=0,1,2 → 3

total 9 = 3×3

tha halving (16) - 16-8-4-2-1 O(log n)

SqrtLoop(16) = $1^2 \leq 16$
$2^2 \leq 16$
$3^2 \leq 16$
$4^2 \leq 16$
$5^2 > 16$ → stop O($\sqrt{n}$)

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③ Second largest

12 - first = 12
35 - first = 35, second = 12
1 - no change
10 - no change
34 - second = 34
1 - no change

answer 34

④ Duplicate check

i = 0 → compare with 1, 2, 3 ...
i = 1 → compare with 2, 3, 4 ... $O(n^2)$

HashSet x = 10 add
x = 20 add
x = 30 add

duplicate x = 20 seen.add(20) → false $O(n)$

⑤ Recursion cost

factorial(4)

4 * fact(3)
3 * fact(2)
2 * fact(1)
1

=>
1
2
6
24
ans 24

⑥ StackAdtDemo

reverse ABC

push A
B
C

Stack : C → Top
B
A

top C
B
A
o/p CBA

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⑦ JavaStackFlow

[A B C]

s.add() - [SNEAKY, A, B, C]

s.remove(B) = [SNEAKY, A, C]

⑧ String Concat Trap

s += "x"

x
xx
xxx
xxxx
...
O(n²)

String Builder append x
append x
append x
...
O(n)

⑨ List Traps

[0 1 2 3 4]

remove(0) = [1 2 3 4]

remove(1) = [2 3 4]

remove(2) = [3 4]

$O(n^2)$

poll() - remove 1st element directly $O(n)$

⑩ Program To Interface

get(0)
get(1)
get(2) $O(1)$

LinkedList easy get(0)
traverse get(1)
traverse get(2)

Total = $O(n^2)$

using next() - Total $O(n)$
↓ cost $O(1)$